ETH zürich

# Mobile Computing

## Specialized Applications
## Consumer and Medical Devices

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# Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

# Overview

- Motivation - why Mobile Computing?
  - Digital therapy management
- The human body and the pancreas
  - Turning food into energy
  - The role of insulin and glucagon hormones
- Insulin Regulation Disorder (IRD)
  - Type 1 and Type 2 Diabetes
- Artificial Pancreas
  - Automated Insulin Delivery
  - Closed Loop Control

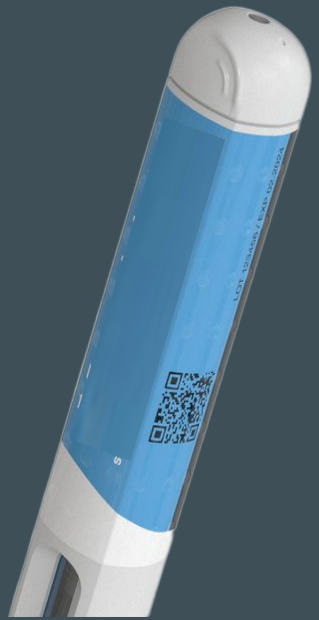


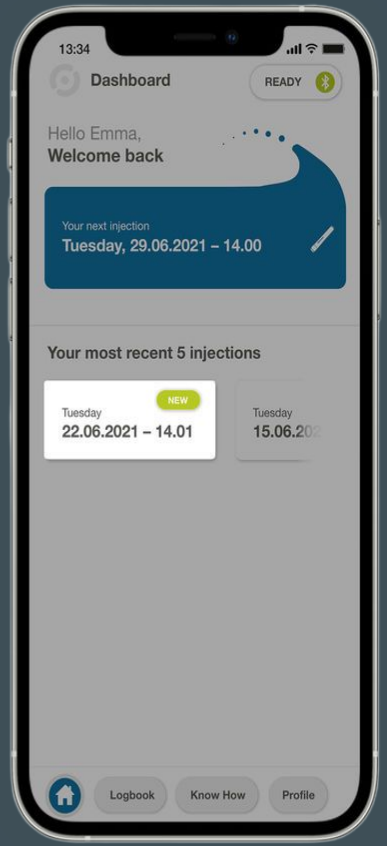
# Mobile Computing for Medical Technology

- Automated medicine delivery is accepted way to administer medication
- Smartphones can augment control systems and offer additional (unlimited) computation
  - Device, edge, and cloud computing
- Therapy management increasingly important
  - Health care professionals in the loop
  - Adherence to a therapy plan, and efficiency can be traced

# Digital Therapy Management

- Wireless data
- VLC for activation





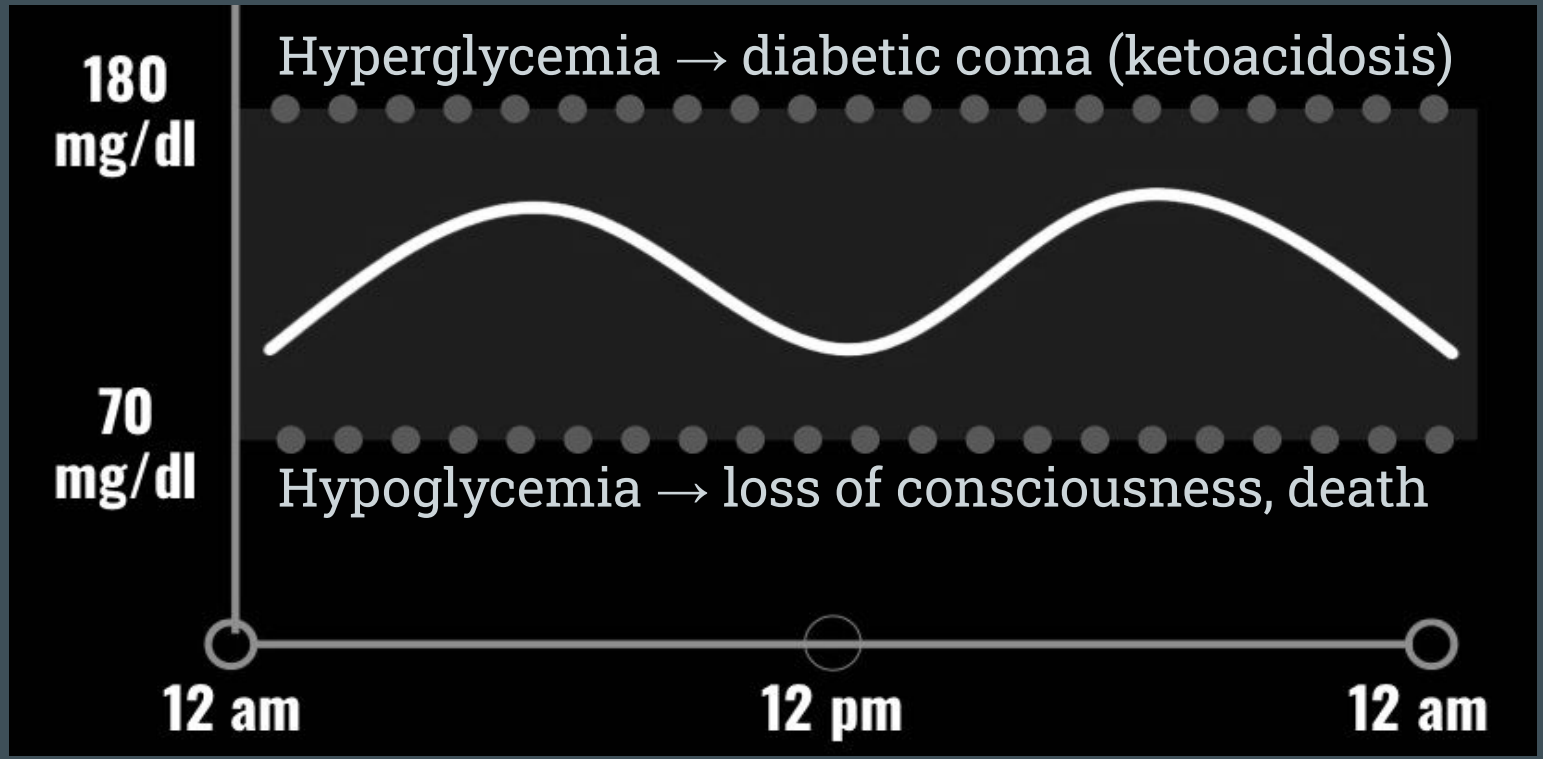
# Digital Therapy Management

- Adherence management of self-medication
- Automated injection logging
- Assist patients, caregivers and healthcare professionals



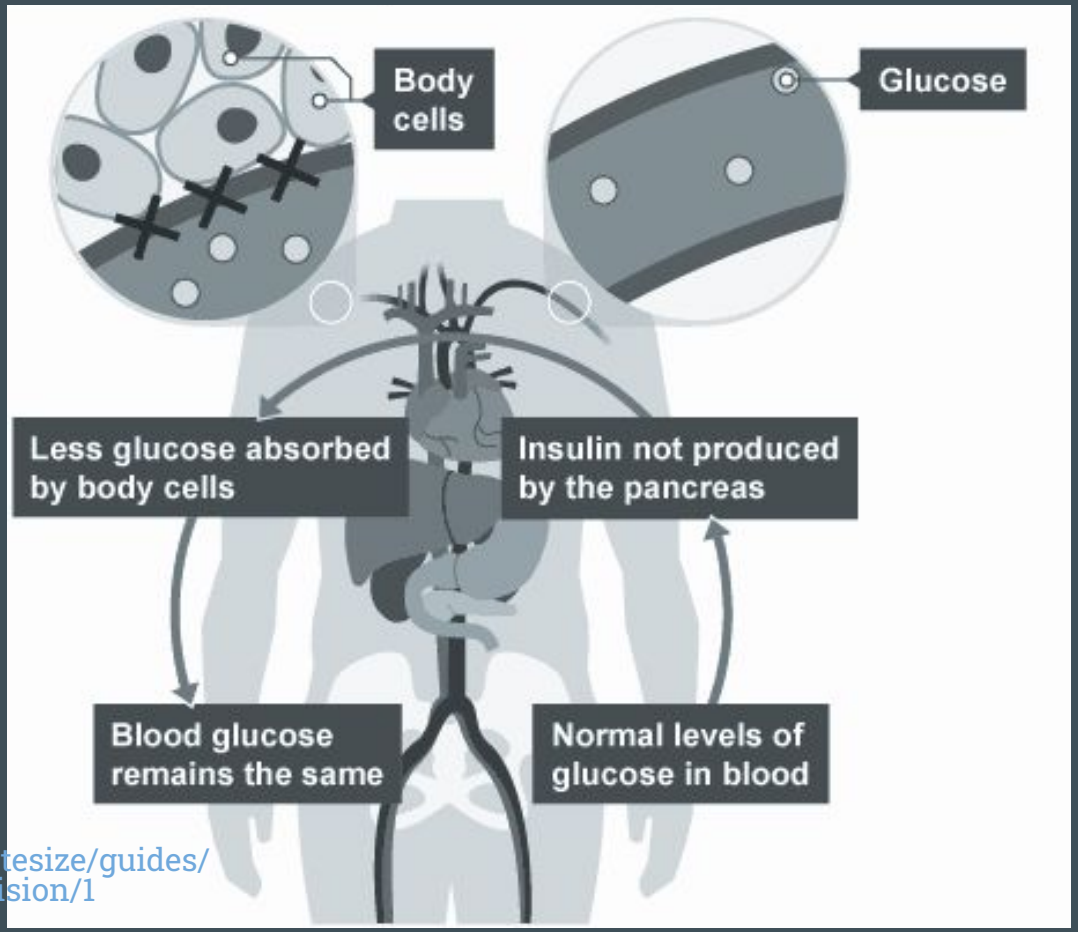
# The Human Body and the Pancreas

- A healthy pancreas regulates the time in range

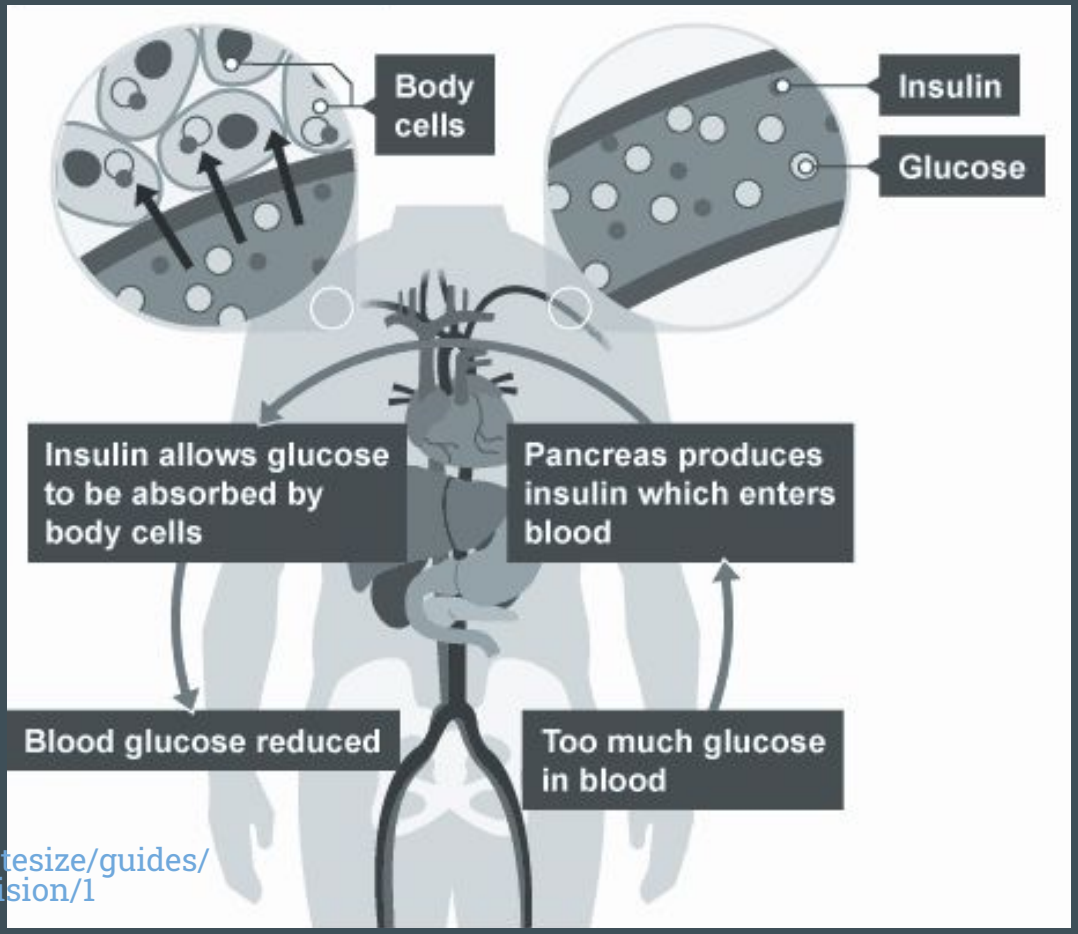


# The Human Body and the Pancreas

- The pancreas (“Bauchspeicheldrüse”) is an organ
- The pancreas generates two hormones: Insulin and Glucagon
  - A hormone is a signaling molecule in a multicellular organism, transported to distant organs to regulate physiology and behavior
  - Insulin helps cells in the body to take up and consume glucose (=sugar) from blood, which lowers blood glucose levels
  - Glucagon causes the liver to release stored glucose, which raises blood glucose levels (= converting glycogen into glucose)




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# Insulin Regulation Disorder (IRD), Diabetes

- A group of metabolic disorders characterized by high blood glucose (hyperglycemia)
  - due to defects in insulin secretion, insulin action, or both
- Considered a chronic, long-lasting health condition
  - affects how a body turns food into energy
  - Type 1: pancreas produces very little or no insulin
  - Type 2: not enough insulin, or body resistance

# Insulin Regulation Disorder (IRD), Diabetes

- Patients with type 1 diabetes and some patients with type 2 diabetes inject insulin to regulate their blood glucose (sugar)
  - Some might inject glucagon as well
- Blood sugar must remain in range
  - to lower risk of long-term complications: blindness, kidney failure, cardiovascular disease

# Living with IRD

- Users have to periodically and manually test their blood glucose levels with a glucose meter, calculate insulin doses
- They administer the necessary insulin doses to lower the blood glucose (3…4 times per day)
  - Bolus: fast-acting insulin, typically taken with a meal
  - Basal: slow-acting insulin to cover the glucose output of the liver, typically taken once a day
- Glucagon injection treats low blood glucose
  - After loss of consciousness (otherwise, just eating sugar)

# Continuous Glucose Monitoring, CGM Sensor

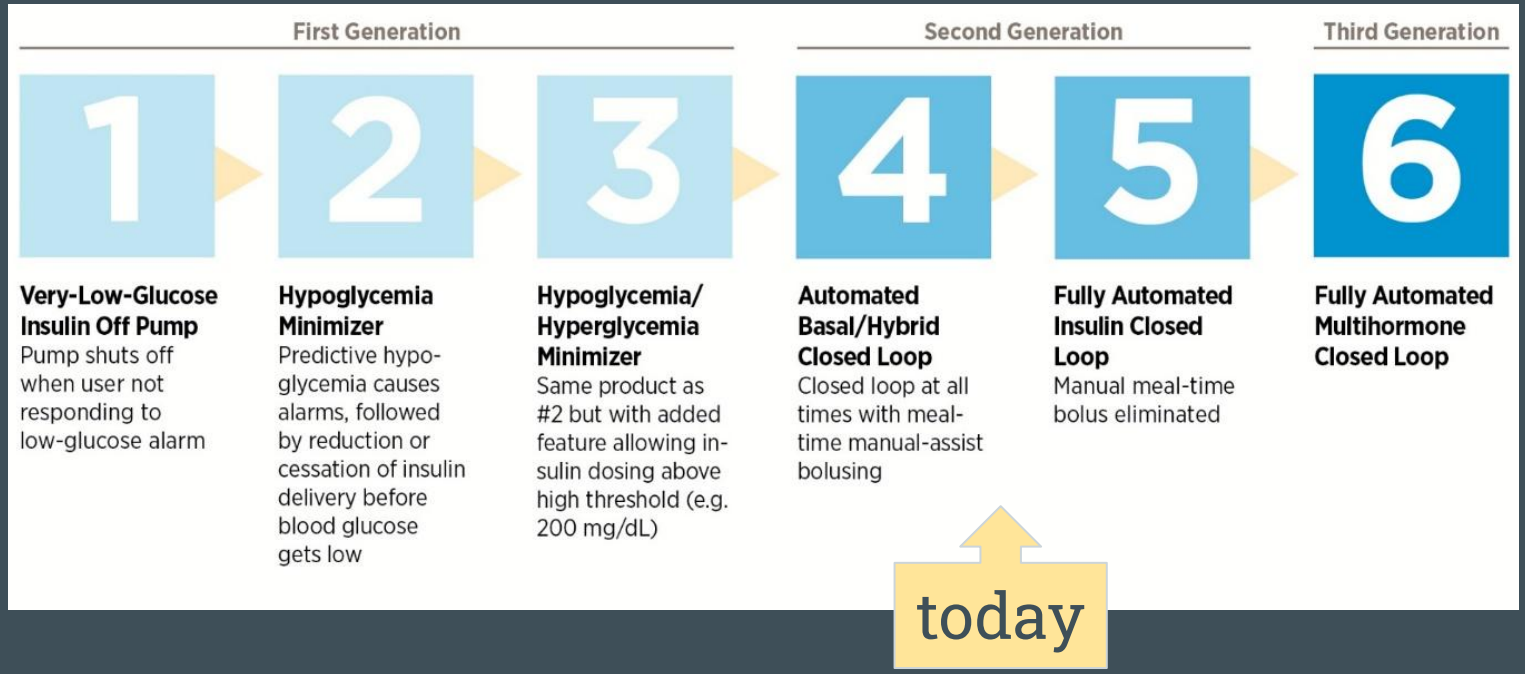


# Artificial Pancreas

- Continuous glucose monitoring system (CGM) & Smartphone
  - Sometimes: Blood glucose meter to calibrate the CGM
  - Insulin infusion pump for subcutaneous (below the skin) injection
- Open-loop: most commercial systems
  - Users read the calculated recommendations on the smartphone
  - Users confirm or change the amount of insulin
  - Users inject their dosis
- Closed-loop: no user interaction, automated delivery

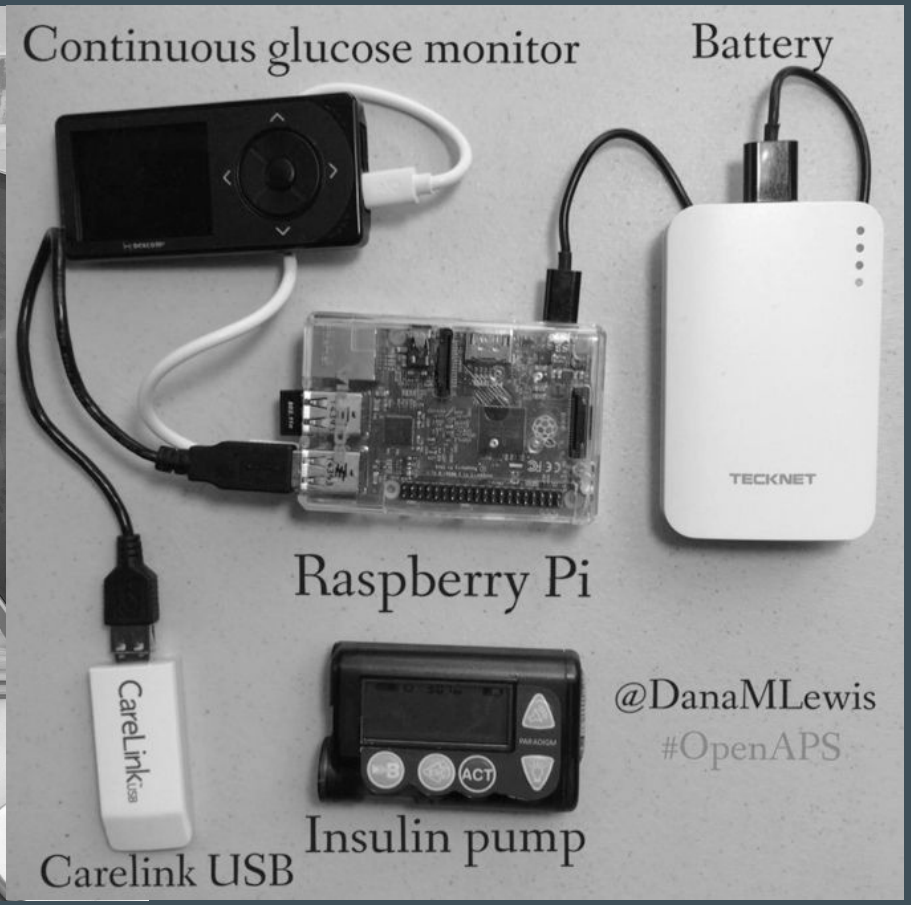
# Generations of Artificial Pancreas

- Growing capability (and complexity) of available systems



# OpenAPS: “Open Artificial Pancreas System”



# Do-it-Yourself System, OpenAPS

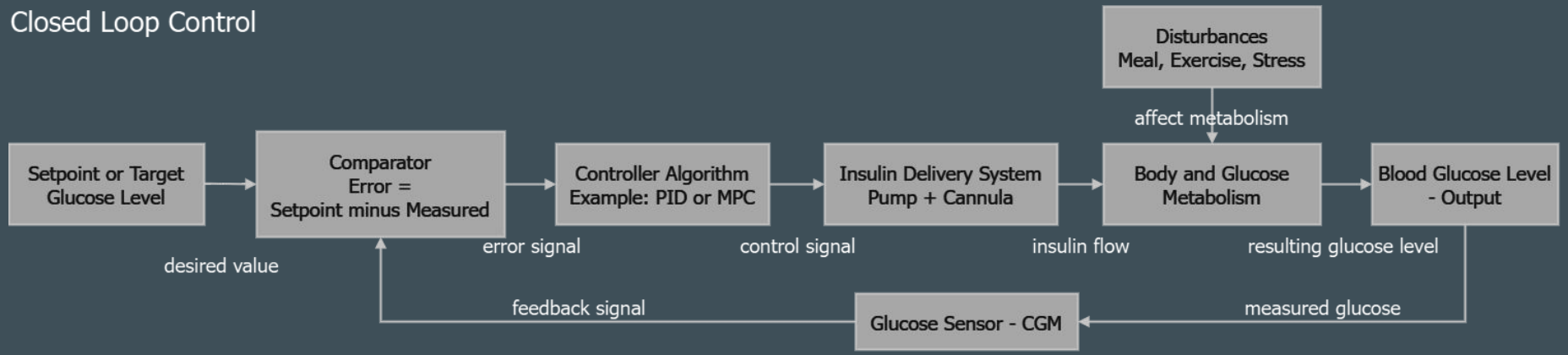


# Control Algorithm (openaps.org/reference-design)

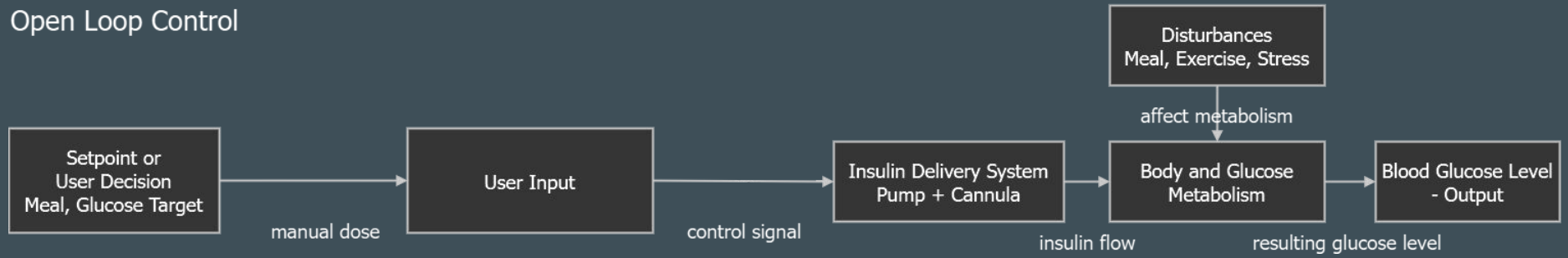
- The OpenAPS algorithm automates insulin delivery with insulin pumps and continuous glucose monitors (CGMs)
- At its core, it relies on a known control approach to manage and predict blood glucose (BG) levels
  - Adjusts insulin dosing based on glucose readings & predictions
  - Keeps blood glucose within target range, if possible
- Designed to be safe and conservative
  - Features like "safety caps" to limit how much delivered at once

# Control Loop

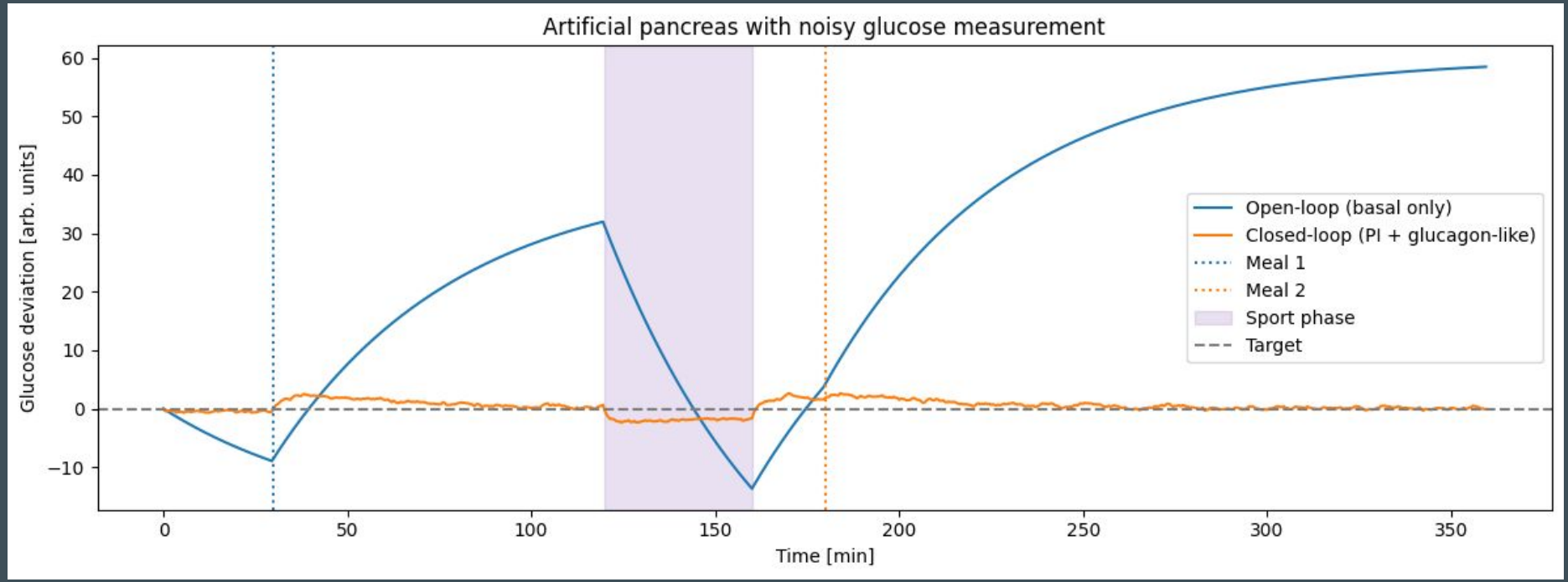
## Closed Loop Control




## Open Loop Control



# Open- vs. Closed-Loop Artificial Pancreas



# Control Algorithm: PID Controller

- The “P” component responds to current deviation between the actual Blood Glucose (BG) level and the target BG level
  - The more significant the difference, the larger the insulin adjustment
  - Only considers the immediate error
  - The P component immediately adjusts insulin delivery if there is a deviation from the target

# Control Algorithm: PID Controller

- The “I” component considers cumulative sum of past deviations over time
  - Addresses sustained deviations that may not be corrected by the proportional component alone
  - If BG levels have been consistently above or below a target, it accounts for the accumulated error & adjusts the insulin delivery

# Control Algorithm: PID Controller

- The “D” component predicts future errors by considering the rate of change of the BG levels (i.e., how fast the BG levels are rising or falling).
  - It prevents overcorrection by slowing down the insulin delivery adjustment if BG levels are changing rapidly.
  - This can help to avoid oscillations or sudden over-corrections that might lead to hypoglycemia (low blood glucose)
  - Helps to smooth out sudden spikes or drops

# Challenges

- Bluetooth 5 security might not be sufficient
  - Might use Ultra Wideband instead
- Reliability of smartphones is difficult to guarantee
  - Battery, background apps, roaming, wireless connectivity
- Augmenting today's deterministic (PID) control algorithms with ML / AI algorithms is not supported by regulators
- 5G connectivity, together with edge computing (computing in the network) might be helpful

# Further Reading

- Background Reading
- wikipedia.org/wiki/Diabetes
- OpenAPS Documentation
- What is an artificial pancreas device system?
- Non-Invasive Glucose Monitoring: Hope or Hype?
- OpenAPS | We moved to freeCodeCamp.org/news
- openaps.org/reference-design

# **Exercise:** Control and Coordination

- Open in your course folder: 02_Notebooks/04_Mobile_Algorithms/04.1_Control_and_Coordination_in_Mobile_Systems.ipynb
- Run the cells and answer the questions.
- Save the completed notebook.

